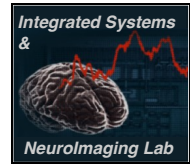


Jiazhen Hong

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PROFILE

Final-year Ph.D. Candidate in Computer Engineering at Rutgers University and Integrated Systems & Neuroimaging Lab

- Research focuses on **machine learning**, **foundation models**, and large language models (**LLMs**) for time-series and neural signals
- Experience spans **brain-computer interfaces**, **biological data analysis**, **multi-agent development**, **large-scale self-supervised model training** on distributed cloud infrastructure

EDUCATION

- Rutgers University, New Brunswick, NJ, USA** *Ph.D. candidate* 01/2020 – Present
Computer Engineering, G.P.A.: 3.9/4.0
Thesis Topic: Artificial Intelligence for Time-Series Signal Processing in Brain-Computer Interfaces (BCIs)
- Stevens Institute of Technology, Hoboken, NJ, USA** *M.Sc.* 09/2017 – 01/2019
Electrical Engineering, G.P.A.: 3.9/4.0
- Jimei & Chung Yuan Christian University, China** *B.Sc. (Double Degree)* 09/2012 – 06/2016
Communication Engineering & Business, G.P.A.: 3.5/4.0 & 4.0/4.0

TECHNICAL SKILLS

- **Machine Learning:** foundation models, self-supervised learning, large language models (LLMs), neural time-series modeling
- **Model Architectures:** Mamba, Swin Transformer, TimeFormer, ResNet, CNN, LSTM, encoder-decoder architectures
- **Frameworks & Libraries:** PyTorch, PyTorch Lightning, TensorFlow, scikit-learn, Hugging Face, NumPy, Pandas
- **Infrastructure & Tools:** Google Cloud Platform (GCP), distributed GPU training, Git, GitHub, Linux/SSH, VSCode, Anaconda
- **Hardware:** Brain Products EEG systems, actiCAP, Raspberry Pi, Arduino

WORKING EXPERIENCE

- Integrated Systems & Neuroimaging Lab, Rutgers University** *Teaching/Research Assistant* 01/2020 – Present
Research
 - Developed ChatBCI, a complete real-time mind-controlled **brain-computer interface (BCI)** speller system enhanced with **LLMs**
 - Led the integration of a **LoRA-Llama** model into the BCI project, incorporating **generative AI**
 - Led and designed an **NSF-funded** temporal perception project involving **neural signal processing** and **Machine Learning**
 - Developed topographic **image representation** for video-like electroencephalography (EEG) signals using **TimeFormer**
 - Created a channel-selection method to improve the efficiency of **time-series** BCI processing for motor-imagery tasks*Teaching*
 - Digital Signal Processing
 - Linear Systems and Signals
- EMOTIV, San Francisco, CA, USA** *Research Scientist Intern* 10/2024 – 09/2025
 - Developed large-scale foundation models for neural signals, trained using distributed pipelines on cloud GPU clusters
 - Designed representation learning frameworks for long-sequence signals, leading to 1st-author publications at **NeurIPS** and **ICDM**.
- Department of Electrical and Computer Engineering, Rutgers University** *Faculty Lecturer* 05/2022 – 07/2022
 - As a faculty lecturer, responsible for teaching a class of 30-40 students in Linear Systems and Signals
- Harvard Medical School** *Student Researcher* 01/2019 – 12/2020
 - Worked on biostatistical analysis of protein biomarkers for **early cancer detection**
 - Applied ML and hypothesis testing (Benjamini-Hochberg correction) to identify significant proteins for cancer prediction

PROFESSIONAL SERVICES & ACTIVITIES

Reviewer

- Conference Neural Information Processing Systems (NeurIPS) 2025 – 2026
- Journal IEEE Transactions on Biomedical Engineering (TBME) 2025
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2023 – 2024

Technical Program Committee

- IEEE International Symposium on Biomedical Imaging (ISBI)
- IEEE International Symposium on Biomedical Imaging (ISBI)

University Services

- Teaching Printer Circuit Board Assembly and soldering for Rutgers ECE department
- Faculty Lecturer for class Linear Systems and Signals

PUBLICATIONS

Journal (Accepted/Published)

- **J. Hong**, W. Qian, Y. Chen, and Y. Zhang, "A geometric approach to k -means," IEEE Transactions on Knowledge and Data Engineering (**TKDE**), accepted, 2025
- **J. Hong**, W. Wang, and L. Najafizadeh, "ChatBCI, a P300 speller BCI with context-driven word prediction leveraging large language models, from concept to evaluation," **Scientific Reports (Nature Portfolio)**, accepted, 2025
- **J. Hong**, P. Rao, W. Wang, and L. Najafizadeh, "ChatBCI-Assist: An Intent-Based P300 Speller with A Locally-Deployed LLM and Adaptive Stopping Strategy Enabling Record Online Spelling Performance," IEEE Transactions on Biomedical Engineering (**TBME**), accepted, 2026, **Invited Submission**

Conference (Accepted/Published)

- **J. Hong**, and L. Najafizadeh "Improving Spelling Performance of LLM-based P300 Speller BCIs using Circular Nonadjacent Flashing and Cross-Validated Calibration," (**EMBC** 2026), **Selected as a Finalist in the Student Paper Competition**
- P. Rao, **J. Hong**, W. Wang, and L. Najafizadeh, "Llama-4-ALS: A Locally Deployable LLM, Tuned with LoRA and Dynamic Hierarchical Data Augmentation, for Speller-Based AAC," (**EMBC** 2026)
- A. Subramanian, S. Mai, **J. Hong**, and L. Najafizadeh, "Neural Dynamics of Time Perception Under Acute Pain: An Auditory Temporal Bisection Study," 23rd IEEE International Symposium on Biomedical Imaging (**ISBI** 2026)
- **J. Hong**, G. Mackellar, S. Ghane, "SpellerSSL: Self-Supervised Learning with P300 Aggregation for Speller BCIs" is accepted by the 39 Annual Conference on Data on the Brain & Mind Workshop @ Neural Information Processing Systems (**NeurIPS** 2025)
- **J. Hong**, G. Mackellar, S. Ghane, "An Efficient Self-Supervised Framework for Long-Sequence EEG Modeling" is accepted by IEEE International Conference on AI for Time Series Workshop @ Data Mining (**ICDM** 2025)
- **J. Hong**, and L. Najafizadeh, "Enhancing Typing Speed in LLM-based P300 Speller BCIs Using A New Data-Driven Dynamic Stopping Strategy," is accepted by IEEE EMBS 12th Annual International Conference on Neural Engineering (**NER** 2025)
- S. Mai, **J. Hong**, T. Shors, and L. Najafizadeh, "Longest Sustained Cortical Activity Duration in Temporal Bisection Task Aligns with Geometric Mean for Log-Spaced Probes," is accepted by IEEE EMBS 12th Annual International Conference on Neural Engineering (**NER** 2025)
- **J. Hong**, P. Rao, W. Wang, S. Chen, and L. Najafizadeh, "ChatBCI4ALS: A High-Performance, LLM-Driven Intent-Based BCI Communication System for Individuals with ALS," 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, (**EMBC** 2025), **Named Top 7% of Submissions**
- **J. Hong** and L. Najafizadeh, "TopoEEG: a TimeSformer-Based Topographic Image Representation Method for Early Single-Trial Detection of P300," 22nd IEEE International Symposium on Biomedical Imaging (**ISBI** 2025)
- **J. Hong**, L. Najafizadeh, "P3T: A Transformer Model for Enhancing Character Recognition Rates in P300 Speller Systems," 58th Annual Asilomar Conference on Signals, Systems, and Computers (**Asilomar** 2024)
- **J. Hong**, F. Shamsi, and L. Najafizadeh, "A Deep Learning Framework Based on Dynamic Channel Selection for Early Classification of Left and Right Hand Motor Imagery Tasks," Proc. of 44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (**EMBC** 2022)

(Under review)

- **J. Hong**, G. Mackellar, S. Ghane, "SAMBA: Toward a Long-Context EEG Foundation Model via Spatial Embedding and Differential Mamba," 2026
- **J. Hong**, S. Mai, and L. Najafizadeh, "CORA: Coordinate-Aware ROI Attention for EEG Representation Learning in Cortical Space," 2026

(In Preparation)

- **J. Hong**, W. Wang, S. Haghani, and L. Najafizadeh, "Subject-specific Channel Selection Based on Davies-Bouldin Index for EEG Motor Imagery Classification," in preparation (Journal)
- **J. Hong**, and L. Najafizadeh, "A Geometric Spatio-Learnable EEG Foundation Model Across Visual and Auditory Modalities in the Temporal Bisection Task" in preparation (Journal)

AWARDS

- ECE Development and Research Excellence Award (2025-2026 cycle), Rutgers University
- Research & Conference Travel Award, funded by the School of Graduate Studies (SGS), Rutgers University
- Trainee Travel Award of 2024-2025 Rutgers Brain Health Institute, Cognitive and Sensory Neuroscience Focus Area
- Travel Award of 2024 IEEE Brain Discovery & Neurotechnology Workshop
- ECE Best Teaching Assistant Award for Fall 2023, Rutgers University